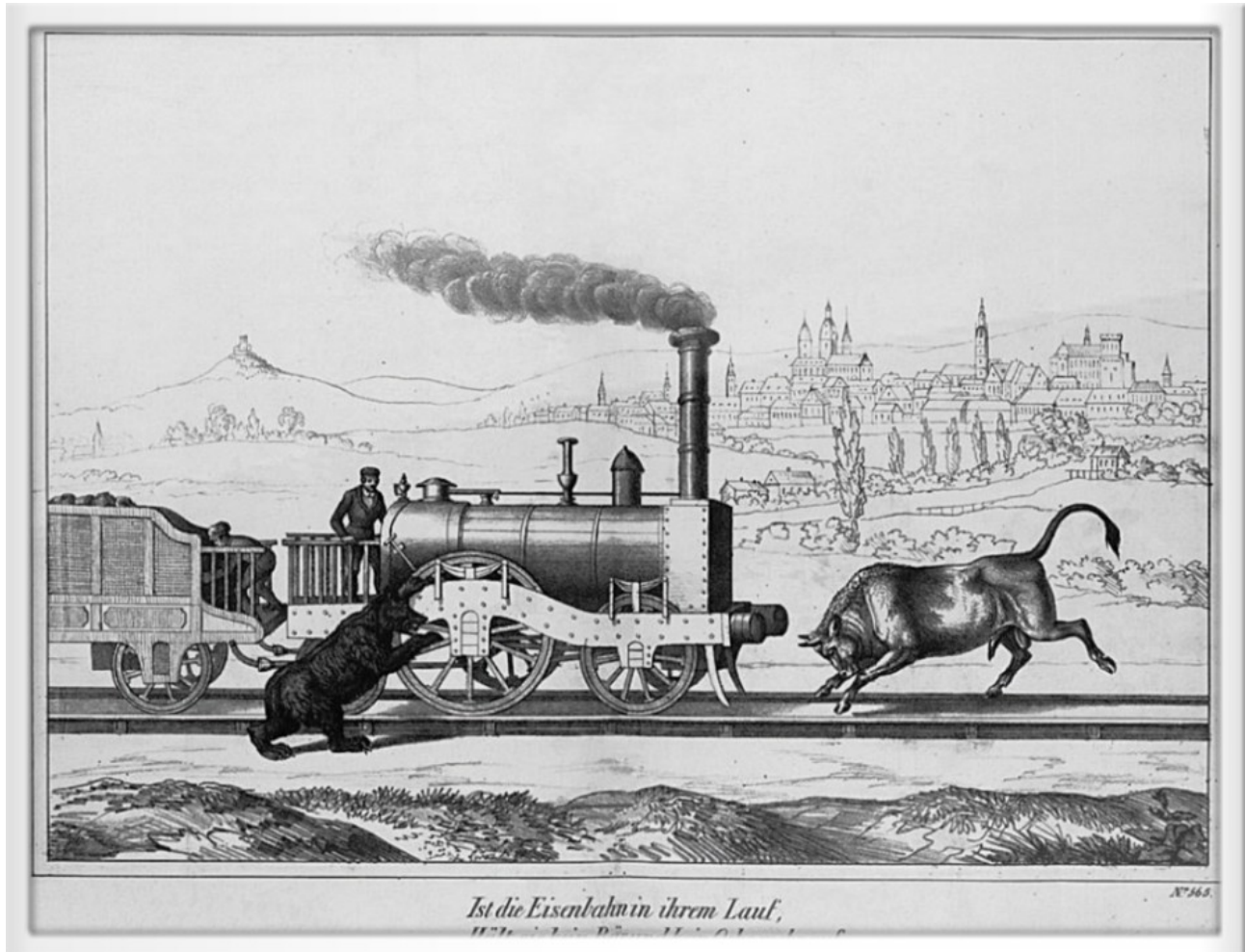


1836 Hesse & Nassau



**"When the railroad is running forward,
no bear or ox will stop it."**

*Caricature from 1844 about the opponents of railroad construction
in the assembly of the Electorate of Hesse-Kassel,
the assembly members Bähr and von Ochs*

1836 Hesse & Nassau

Railroading in central Germany – an 18xx game for 3 to 5 players

**by Karsten Ockenfels and Dirk Steiner
based on an gaming idea of Francis Tresham**

Rulebook

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1. Introduction

During spring in 2001, I was in my final year at the University of Marburg and my bride in her first pro year in Korbach. Several times we went by car from Marburg to Korbach along Eder river and I noticed a small railway from Frankenberg to Korbach with a beautiful landscape around. But my bride told me the railway wasn't used any more. I was sorry about that and thought this track should be in an 18xy version for Hessen. This short idea was a first step in a long development of this game. For some German regions such regional variants already were published by Wolfram Janich. Now from time to time I brainstormed for a regional variant for Hessen. Two years later at Easter days in 2003 I visited Bjoern Rabenstein in Berlin for some days of gaming. In his apartment I found a map for Germany showing all important railway tracks. I took a piece of paper and started sketching a map for my 18 Hessen with some company ideas. Development officially had started. Playtesting was mainly done by 18xy groups in Marburg and Berlin with inputs from groups in Munich and Hamburg. After only one play test during evening before German 1835 championship tournament in Marburg in 2003 version 1 was skipped and changed to version 2. This one had some quite successful tests, but it failed a legendary test on Easter days 2006, when I was in Berlin again. One of the players called „Fax“ tried a radical strategy using concessions proving the version 2 concept of concessions to be dysfunctional. On top of that some of the 13 original privates were just for the flair rather than useful for the game. Generally this version was overloaded with too many ideas and specialties, so we had to sort out several things. During the following months version 3 was developed. which was already not far from today's version in basics, companies and privates. We tested this version with several games for about two years and were pleased with the concept. But we found problems in game balance because of the technical structure of this version. In this situation probably happening to many games during development my friend Dirk Steiner took over. He had correct analysis of problems and the right ideas, where I was just loving my original ideas or where I just didn't see solutions. We improved companies and privates and reformed the map, which wasn't updated since version 1. And Dirk developed the SNOW-system for transit routes. By the time I had read and worked with a doctoral dissertation about early railways in the regional states of Hessen and Nassau by Ludwig Brake. That gave me an option to introduce some small changes to make the game more historical. The result was version 4 in 2009. For several years we were very pleased with it and thought about publishment. But Dirk took it for a final test to a convention in Munich. Munich's 18xy group gave one serious criticism: There were two small companies called Lahn & Eder Railway and Waldeck's Local Railway merging into one company later in the game. Munich players thought this to be too many special rules and too much complication for one small element by the cost of game balance. They were right, so I sadly had to bid farewell to the non-historic Lahn & Eder Railway, whose track had once initiated my thoughts about the game in 2001. Instead WLB - being my declaration of love for the small country of Waldeck – became a normal big company. That's why it is the only non-historic company in the game. Now we had version 5. Tests for that were successful over the years, some details were adjusted resulting in version 5.5, which you now have got. We think this version works. Additionally we found it to be a useful for learning games with 18xy rookies. I have to admit this never was my intention, but I like the outcome. My sincerely thanks go to all the players testing the game over the years, especially to Georg Brede and Bjoern Rabenstein for all the time. My special thanks go to Dirk Steiner for his extensive input in game development and to our friend Antje Laurson for designing the map. Also I want to thank Ludwig Brake for his book about railway development in Hesse and Nassau. We hope you'll have fun with 18HN.

Karsten Ockenfels, April 2016

2. Game parts

The following components should be available:

- 1 map including share value table
- 1 game phase overview at the end of this rule book
- 68 yellow track tiles
- 49 green track tiles
- 38 brown track tiles
- 6 bridge markers
- 5 position cards for turn order
- 1 priority deal card
- 35 train cards
- 90 share cards
- 8 cards for private companies
- 5 cards for concession rights
- 40 concession cards
- 10 small cards to show obligated track building („OK“)
- 50 station markers and share value markers (5 per company)
- 1 game phase marker
- 10 company sheets
- this rule book

You should add paper money or poker chips in sum of 12000 Mark (German currency) as the bank.

3. Game preparation

Before a game, we recommend to sort track tiles, bridges and company markers around the map, to put company shares into their boxes on the map with director's share on top and to put reserved shares aside for now. The trains should be put in their boxes on the map. Concession papers can be stacked with their concession right cards near the matching private. One player manages the bank.

Draw the order of players for initial auction. Give the players their initial money:

3 player game 800 Mark each

4 player game 600 Mark each

5 player game 500 Mark each

This money is taken from the 12000 Mark in the bank.

Put the seven private company cards on the table in their order (see chapter 6).

All players should have enough space on the table to put their money, shares and company sheets in front of them. If you play with a digital moderator you'll only need track tiles, bridges and markers.

Players should agree before game start about using optional rules, for example which system for determination of player order during share rounds they use or if they want to use the additional private resulting in additional 10 Mark starting money.

We recommend also to talk about house rules etc. before game start and to use time when other players are acting to think about your own plans. This will speed up the game.

Players' money and company sheets are always open to all.

Deals between players are not binding but should be treated as gentlemen's agreements.

Negotiations are always conducted openly.

4. General gameplay and victory conditions

1836 Hesse & Nassau („18HN“) is an economic game of railway development for 3 to 5 players. It is part of the family of 18xy games covering the region of central Germany called Hesse today. In this game, players try to make the biggest wealth out of their starting money combining cash and value of their shares at the end of the game. Depending on turn phase players act as investors selling and buying shares at the stock exchange during stock rounds (SR) and they act as directors of railway companies deciding their operations during operation rounds (OR). Players gain profit from fixed income of privates, eventual dividends from company shares when they pay out earnings and from value raises of their shares.

A game therefore alternates between stock rounds (SR), when players do their share trading, and operation rounds (OR), when companies operate by decisions of their directors. Depending on game phase, between stock rounds there will be one, two or three operation rounds: one OR during so called yellow phase, two OR during so called green phase and three during so called brown phase. This number of OR will be determined at start of preceeding SR. At start of a game it is in yellow phase. Green phase starts when the first type-3-train is bought, brown phase starts when the first type-5-train is bought (see 8.4.6). A group of one stock round and its following operation rounds is called a turn set or shortly „set“.

A game finishes at the end of a set, in which the bank runs out of money at least once no matter if during SR or OR. But payouts continue normally until the game is finished even if bank goes into negativ finances. The complete set will be played to the end. As another option a game ends immediatly when a player goes bankrupt (see 8.4.7). After end of game players finally calculate their wealth summing up their final cash and final values of their shares. Share values can be seen on the share value table. In a rare case of still existing privates they count their original face value. The player with the highest wealth at end of game will be the winner. If the game ends by a player's bankruptcy this player will automatically take the last place.

5. Companies, concessions und privates

Shares and papers in this game have different functions. There are privates, concession rights and company shares. When buying a private or a concession right, a players gets 100% of it on one paper. Shares are worth 10% or 20% of a company. Please note the following details:

5.1 Companies

Each company represents one railway corporation, which can operate on the map. Companies lay track, claim stations and buy trains, which enable a company to use routes on the map that make income for the company. That income can be withhelded into the company's treasury or be paid out to the share holders (see chapter 8). Companies enter the game sooner or later by players' choices. Each company is managed by one of the players, who alone as the company's director decides about the company's operations. Company shares are traded in stock rounds (SR).

5.2 Concession rights

At the time of this game there were several small independent states in the region, that contains the German federal state of Hesse and the east of the German federal state of Rhineland-Palatine today. In mid 19th century there were mainly four bigger states in the rank of duchies: Hesse-Kassel, Hesse-Darmstadt, Nassau and Waldeck. Also there were some smaller areas like the free city Frankfurt am Main, the county of Homburg und the city of Wetzlar belonging to Prussia. There are concession rights for the passage of Frankfurt and for the four bigger areas administring permission to do railway operations in the area (see 8.4.1). These five concession rights are never traded. They are connected to privates during starting auction (see chapter 6). Concession rights don't count against paper limit. They lose their significance at start of brown phase, which represents the time when the German empire was founded, skipping independence and borders of small German states.

5.3 Privates

Privates are small companies, which enter the game during the initial auction. A player who buys a private will own 100% of this small company until its closure. A private's economic activity only involves a regular income for its owner for the duration of its existence and an execution of a special ability at a defined moment during the game (see 8.2).

Four of the privates will be exchanged for a reserved share of a certain company during the game when they execute their special ability. To do that take the corresponding reserved company share in exchange for the private's paper after executing the special ability.

The other three privates (or four when adding the optional private) will close at start of brown phase or in some cases even earlier if their special ability says so. In this moment remove the private's paper from the game without compensation. A closed private doesn't count against share limit any more. There are seven privates for a standard game in a fixed order for the initial auction:

1. Homburg's Railway („Homburger Bahn“) face value: 20, income: 5
special ability: It closes at start of brown phase,
in this moment only, the owner immediately may – not must – lay the brown tile for Bad Homburg as a special tile lay on the green printed Bad Homburg hex without paying the hex's building cost.
If the owner decides to decline this any company may lay the track later paying 40 M.
2. Building Company Rhine Bridges („Baugesellschaft Rheinbrücken“) face value: 40, income: 10
special ability: It closes at start of brown phase at the latest.
During yellow phase and green phase any company, whose director owns this private, is allowed to build one bridge over Rhein river or estuary of Main (the blue borders in the west) per turn for free and in addition to the normal track lay (before or afterwards). Such a bridge must extend the building company's reachable track. It is possible to use this ability with two different companies in the same OR as long as the private's owner is director of both companies.
When building the first bridge, flip the private to its back side without changing its income.
When building the second bridge, close the private before start of brown phase.
3. Frankfurt's Local Railway Co. („Frankfurter Lokalbahnen AG“) face value: 50
income: 10 times the made tile lay (yellow or green) in both Frankfurt hexes, so max. 40
special ability: At start of each OR this private can build or upgrade one of the two hexes of Frankfurt city (green track not before green phase of course). That's a maximum of 4 tile lays.
Before brown phase the companies are not allowed to build in Frankfurt. At start of brown phase this private closes. Afterwards all companies can build or upgrade the Frankfurt hexes.
The owner of this private also possesses the concession right for the passage of Frankfurt.
As long as this private isn't closed, each company can buy such a concession once per game paying 40 M to this private's owner. For the rest of the game this concession allows a company to use the special track between Frankfurt's western station and eastern station for a route (see 8.4.5), in which case Frankfurt's two stations count as one station with the higher station value (if different). This concession can't be bought any more after this private has closed.
4. Building Company Eder Valley Dam („Baugesellschaft Edertalsperre“) face value: 60, income: 15
special ability: It can be exchanged for 10% WLB during green phase at start of any OR or SR.
In this moment, the owner must build a yellow tile in Bad Wildungen even without connection to other track as an additional tile lay for free. At the end of an OR during which Bad Wildungen has been built otherwise or in the moment when brown phase starts this exchange must happen (including the special yellow tile lay if not already happened), even if WLB hasn't been founded yet. If the exchange happens during an OR the new WLB share won't pay WLB dividend this OR. If WLB floats because of the exchange it will operate in the following OR for the first time.
This private's owner also possesses the concession right for Waldeck (see 8.4.1).

5. Frankfurt & Hanau Railway („Frankfurt-Hanauer-Bahn“) face value: 70, income: 15
 special ability: It can be exchanged for 10% FHB during green phase at start of any OR or SR.
 In this moment, FHB gets a second home station in Hanau for free, even if Hanau isn't built yet.
 In the moment when brown phase starts, this exchange must happen (including the special token) even if FHB hasn't been founded yet. Therefore one station in Hanau always is reserved for FHB.
 If the exchange happens during an OR the new FHB share won't pay FHB dividend this OR.
 If FHB floats because of the exchange it will operate in the following OR for the first time.
 This private's owner also possesses the concession right for Hessen-Kassel (see 8.4.1).
6. Taunus Railway („Taunusbahn“) face value: 80, income: 15
 special ability: It can be exchanged for 10% WEG during green phase at start of any OR or SR.
 In this moment, the owner must build a yellow tile in a Taunus hex (T), even without connection to other track as an additional tile lay for free. At the end of an OR, during which all 3 T-hexes have been built otherwise or in the moment when brown phase starts, this exchange must happen (including the special yellow tile lay if not already happened), even if WEG hasn't been founded yet. If the exchange happens during an OR the new WEG share won't pay WEG dividend this OR.
 If WEG floats because of the exchange it will operate in the following OR for the first time.
 This private's owner also possesses the concession right for Nassau (see 8.4.1).
7. Odenwald Railway („Odenwaldbahn“) face value: 90, income: 15
 special ability: It can be exchanged for 10% SB during green phase at start of any OR or SR. In this moment, the owner must build a yellow tile in an Odenwald hex (O), even without connection to other track as an additional tile lay for free. At the end of an OR, during which all 3 O-hexes have been built otherwise or in the moment when brown phase starts, this exchange must happen (including the special yellow tile lay if not already happened), even if SB hasn't been founded yet. If the exchange happens during an OR the new SB share won't pay SB dividend this OR.
 If SB floats because of the exchange it will operate in the following OR for the first time.
 This private's owner also possesses the concession right for Hessen-Darmstadt (see 8.4.1).

For the optional additional private see chapter 9.

If you play with it, it will be auctioned as the second one before Building Company Rhine Bridges.

6. Initial auction

First, randomly determine player order for the initial auction. You can use the position cards to do that or use your own method. Players receive their starting money from the bank depending on the number of players. The game starts with the initial auction of the privates. Put the seven privates in their order as described in chapter 5 and put the five concession rights to their corresponding privates. If you play with the optional private „Seidler & Siebrecht“ put it between „Homburg's Railway“ and „Building Company Rhine Bridges“ and give every player 10 M more from the bank. The privates will now be auctioned in this order, one after another.

In determined player order, the players use their starting money to bid for the current private. The opening bid on a private must be at least its face value, although it may be higher already. Every following bid must be higher than the preceeding one. All bids must be multiples of 5. When a player doesn't want to make a higher bid or any bid at all on a private, he drops out of the auction of this private. Bidding for a private continues in player order until all but one player have left the auction. The player with the highest bid pays that money to the bank and receives this private plus the corresponding concession right if any. Then the auction for the next private starts with all players. Player 1 starts the auction of the first private. As soon as someone has bought it, player 2 will start the auction of the second private and so on. Note that the player, who starts the auction of a private, is allowed to leave the auction directly without making an opening bid.

Theoretically, a situation can occur where nobody wants to make an opening bid on a private.

If all players decline to make an opening bid of at least 20 M on Homburg's Railway the auction of this first private will restart with player 1, but the minimum bid for it will be decreased by 5 M to 15 M. If this happens again, the minimum bid for it will be decreased to 10 M, then to 5 M, then to 0 M. If the minimum bid reaches 0 M, player 1 will be forced to bid 0 M as an opening bid.

If all players decline to make an opening bid on any other private the initial auction will be stopped for a short operation round, because some privates already have been bought. If „Frankfurt's Local Railway Co.“ already has been bought it can lay track in a Frankfurt hex in this short OR. The privates pay their income to their owners now. Afterwards, continue initial auction with the private that didn't get an opening bid before. In this case an opening bid still must match face value.

The initial auction is finished, when all privates have been bought.

Then a new player order is determined, which will be in effect for the rest of the game.

The player with the lowest amount of cash after the initial auction is the new player 1, the player with the second lowest amount of cash is the new player 2 and so on. In case of a tie use the old player order of the initial auction as tiebreaker. Give the priority deal to the new player 1. He will start the first stock round. For alternative methods of the initial auction see chapter 9.

7. Stock rounds (SR)

7.1 Players' options during SR and share limits

During stock rounds players do their actions clockwise in player order until the stock round ends. Note that the player order has been determined for the game after the initial auction. The player holding the priority deal starts an SR. A player's action involves at least one trade or he/she passes. To trade stock a player may first sell any number of shares respecting the restrictions for stock trading (see below), then the player may buy one share. The player is allowed to sell only or to buy only, but doing neither of both means that the player passes.

A player that passes isn't automatically out of the SR. As long as another player does a trade afterwards, a player, who passed earlier, will get another option to trade in player order. Only if all players pass consecutively in player order the current SR will end. On the other hand, a player taking an action other than passing is guaranteed another stock turn before this SR will end. When all players pass consecutively, the priority deal for the next SR goes to the player positioned after the last trading player in player order. You could also say: the first passing player of the consecutive passing will start next SR, so stock trading is just paused for some operation rounds. To mark the priority deal use the priority deal card or for more flair a wooden train model or something similar. For alternative methods to determining next SR's player order see chapter 9.

Stock trading never happens between players directly. Player always do their trades with the bank. The only stock that can be traded during stock rounds are the shares of the ten companies. Players can't trade privates or concessions during stock rounds. Note that there are two different positions for shares belonging to the bank: They can still be in the initial offering („init shares“) or already in the open market („pool shares“).

When a player sells shares, he sells them to the open market receiving the current value of the sold shares in cash from the bank in most cases. Only if sold shares are part of a company that hasn't operated yet the player will only receive the new lower share value in cash (see 7.2).

When a player buys a share, he can choose to buy either an init share or a pool share as long as both are available. In the beginning, all shares are in the initial offering, which is represented by the company boxes on the map. These init shares always cost their par value (see 7.3) instead of the current market value. Pool shares always cost their current market value. Share values count for a normal share representing 10% of a company. Director's shares representing 20% are worth double share value.

Some limitations must be respected for trade actions:

A player never is allowed to own more shares than stated by the share limit. The share limit depends on the number of players: 28 shares in a 3-player-game, 21 shares in a 4-player-game, 17 shares in a 5-player-game. Sometimes, a situation can occur where a player suddenly has too many shares. In this case, the player is forced to correct this by selling shares on his next opportunity.

The first share of a company bought by a player from the initial offering must be the director's share (see 7.3). Furthermore a player must not buy init shares of a company, when the player already owns at least 60% or more of this company. For this limit do not count a share that the player will receive by exchanging a private later in the game. Such a limitation for buying a share doesn't exist for shares in the open market or for private exchange, so a player can own up to 100% of a company. But you must not buy shares of companies whose shares you earlier have sold during same SR.

Selling shares to the open market is possible, as long as the open market won't hold more than 50% of a company. For this limit do not count shares that still are in the initial offering.

Furthermore shares must not be sold before their company has floated (see 7.3).

To sell a director's share there must be another player who immediately can become the new director. This means that this player must own at least 20% of the company (see 7.4).

In the first stock round of a game, it is forbidden to sell any shares.

7.2 Consequences of share trading for share values

Every company, that already has been founded, has a share value marker in the market table. This marker shows the company's current share value. During a stock round, a company's share value is adjusted one step downwards on the market table each time a player sells any number of shares of this company in one trade action. If a player sells several shares of a company in several different trade actions the share value will decrease one step every time, but of course the player receives only the current share value for every selling. If the share value marker already is on the lowest step of a column it won't decrease further by selling. Shares of a founded but not yet floated company can't be sold, although they already have a share value marker. If the company has floated but not yet operated adjust the share value marker before calculating the money the player gets for selling.

At the end of an SR, all companies that are sold out (no init share and no pool shares) will get a share value adjustment of one step upwards if the marker is not in the upmost row already. Do that in OR order if necessary. For this purpose ignore the reserved shares for unexchanged privates.

Any time a share value marker reaches a position on the marker table, where already markers of other companies are, put the moving marker under the other ones. This is important for OR order.

7.3 Founding and floating a company

A player founds a new company by buying its director's share from initial offering during an SR. This player becomes the company's director and chooses its par value, which can be 70, 75, 80, 85, 90 or 100. The player pays doubled par value to the bank for the director's share, because it is a 20%-share. Put one company's marker in the par value box to mark the par value for the rest of the game. Put another marker on the market table into the darker coloured share value box, which has the share value corresponding to the par value. Now the company is in play and all players are allowed to buy 10%-init-shares of the company for par value. As soon as the players hold shares in sum of 50% of the company, the company floats. It is possible, that a company floats at the start of an OR, because a player exchanges a private for the reserved share of this company and during the SR exactly 40% have been bought, so the exchange results in 50%. In this cases the company is considered floated for this OR (see 8.1). If a company floats during an OR as consequence of a forced private exchange, when brown phase starts, the company will operate for the first time in the next OR. Note that players can only sell shares of floated companies and only floated companies operate during OR, so a company's share value can't change before it floats.

When a company floats, put the company sheet in front of its director and put the station markers on

the company sheet into their circles. Then give the company its starting money from the bank. A company receives 5 times its par value when it floats. As soon as the company's track obligation is fulfilled, it receives the same amount of money (5 times par value) again (see 8.4.3). So a company's capitalization is done in two parts. That's why it is important to mark a company's par value. As an exception to this rule a company will receive its full capitalization of ten times the par value for its floating immediately if it is founded (not floated!) after the start of brown phase. After floating, a company will operate in every following OR.

7.4 Change of directorship

From the moment a company is founded until the end of the game, there always must be a director for the company. Usually this is the player, who has founded the company. But it is possible, that directorship changes to another player during the game. This can happen in two different ways:

One option is a takeover by another player, who collects more percentage of the company than the current director. The other player can do that by buying enough shares or by exchanging a corresponding private to collect enough shares. It is not enough for a takeover to have as many percentage as the current director, you need to surpass him. That means, such a takeover won't be possible if a director owns at least 50% of the company. When a takeover happens, the new director gets the director's share from the former director in exchange for two 10%-shares. The company sheet with all its assets is put in front of the new director.

As the other option, the current director can sell enough shares of the company during an SR to own a lower percentage than another player because of this selling. Thereby the current director wants to transfer his directorship to another player. But to do this there must be another player who owns at least 20% of the company in this moment, because a new director must be able to hold the director's share, which represents 20%. So a player can't become the new director, as long as he doesn't hold more than 10%. If several players are qualified to become the new director by owning the same percentage the new director will be the one of them who is closest behind the former director in player order (see example). Exchange shares and transfer the company sheet as described before. Remember: To transfer directorship by selling shares you must respect the limitations for trade actions (see 7.1). The second option can also occur during an OR, when a director is forced into an emergency money raising (see 8.4.7).

Example: Aragorn owns 20% VB, Bilbo is VB-Director owning 50%, Denethor owns 10%, Eowyn owns 20%. It's Bilbo's turn during SR and he sells 40% VB, so share value decreases one step. Now there are players who hold more percentage of VB. Eowyn becomes the new director. She and Aragorn both own 20%, which is the highest percentage now, but Eowyn is earlier than Aragorn in player order in this moment. If Denethor as well sells 10% VB now, decreasing share value another step, there will be 50% in pool and no more shares of VB can be sold until somebody buys a pool share of VB first. Otherwise Eowyn gets the option to sell 10% of VB to make Aragorn the new director. If Eowyn neither sells nor buys a share of VB in her next action, Aragorn will get the option to takeover VB by buying a share, because in this case he will hold 30%.

7.5 Ending a stock round

When all players consecutively pass their action during an SR, this SR is finished.

Before the following OR starts, there is some short administration to do:

Determine the player order for the next SR as described in section 7.1.

Adjust the share values of sold out companies one step upwards if possible (see 7.2).

Check again how many operation rounds will follow: 1 or 2 or 3 (see 8.1).

If the game is currently in green phase, the owners of the four privates that can be exchanged for the corresponding reserved share get the option to do so now.

Such an exchange eventually could float a company (see 7.3 and 5.3).

8. Operation rounds (OR)

8.1 Number of operation rounds and exchange of privates

The number of operation rounds between two stock rounds is determined at the start of the previous SR and depends on the game phase. During „yellow phase“ there is one OR per set, during „green phase“ there are two ORs per set, during „brown phase“ there are three ORs per set. If the game phase changes within an OR, the current number of ORs for the current set won't change.

During „green phase“ there is a short intermediate round before each OR and SR. In this intermediate round the respective owners of the four exchangeable privates must decide whether to keep the private receiving its income at start of OR or whether to exchange the private for its corresponding reserved share, which includes immediate exercise of the private's special ability. These decisions are made in the order of the still existing privates corresponding to initial auction.

By exchanging a private during the intermediate round a company eventually can be floated. This will be the case if before the exchange 40% of the company are owned by players and because of the exchange this increases to 50%. In this case the company will already operate in the upcoming OR. Note that this can't happen during „yellow phase“, because private exchanges don't happen then.

Private exchanges are forced to happen at start of brown phase at the latest. Companies that float in that moment – if any – operate in the following OR for the first time.

8.2 Operating privates

First in an OR, the remaining privates take their turn in private order corresponding to initial auction. For all remaining privates except the Frankfurt's Local Railway Co. the bank pays the income to the owners. The special abilities of these privates are exercised at another time (see 5.3).

The Frankfurt's Local Railway Co. eventually may lay track in one of the two Frankfurt hexes before its owner gets the current income from the bank.

8.3 Order of companies during OR

Each company operates exactly once per OR doing all parts of operation. The order of the companies begins after a possible intermediate round and private operations, because companies still could float by exchanging privates during green phase. Otherwise the order of companies only can change within an OR, when shares are sold during an emergency money raising (see 8.4.7). Company order depends on current share values of the companies. Companies with higher share values operate before companies with lower share values. If companies have the same share value but in different boxes on the market table, companies positioned further to the right operate before companies positioned further to the left. Companies in the same box are ordered from top to bottom according to the marker stack. That's why it is important to put a share value marker under the other ones when reaching a new share value box. After each company the next operating company is determined that way. After the last company has operated in that way, the OR is finished (see 8.5).

8.4 Operating a company

When a company operates for the first time after floating, it begins by placing its home station marker on its starting space as a special action. If there is already a track tile on this hex, one station marker position must be reserved for this home marker. A company's OR then follows a specified sequence: lay or upgrade track (8.4.2), place a station marker (8.4.4), run trains, calculate revenue and declare dividends (8.4.5), purchase trains (8.4.6). This sequence can't be scrambled. Additionally a company can purchase concessions at any time during this sequence (8.4.1) and track route obligations of the companies can be fulfilled (8.4.3).

The following sub-chapters describe these parts of a company's OR in detail.

8.4.1 Concessions and companies

During „yellow phase“ and „green phase“, a company can purchase concessions at any time during its OR. The company may purchase more than one concession within the same OR, but only one concession per country and not the concession for its home country since this is granted to the company automatically. Acquisition of a corresponding concession is the prerequisite for operating in this country in any way. Therefore the company must have the concession before operating there, but the concession counts for the rest of the game. This even applies to the completion of the obligated route or to the use of already laid track in the country to reach hexes in another country. The concession called right of passage in Frankfurt allows to use the special track in Frankfurt hexes (see 5.3). To purchase a concession a company pays 40 Mark to the player who owns the corresponding concession right, which has been auctioned with the privates. Each company has its home concession automatically for free when floating. The companies' home countries are:

Principality of Waldeck („Fuerstentum Waldeck“):

Waldeck's Local Railway („Waldeck'sche Landesbahn“) - WLB

Electorate of Hesse-Kassel („Kurfuerstentum Hessen-Kassel“):

Friedrich Wilhelm Northern Railway („Friedrich-Wilhelm-Nordbahn“) - FWN

Main & Weser Railway („Main-Weser-Bahn“) - MWB

Fulda & Hanau Railways („Fulda-Hanauer Bahn“) - FHB

Grand duchy of Hesse-Darmstadt („Großherzogtum Hessen-Darmstadt“):

Main & Neckar Railway („Main-Neckar-Bahn“) - MNB

Starkenburger's Railway („Starkenburger Bahn“) - SB

Hessian Louis Railway („Hessische Ludwigseisenbahn“) - HLB

Vogelsberg Railway („Vogelsbergbahn“) - VB

Duchy of Nassau („Herzogtum Nassau“):

Wiesbaden's Railway Company („Wiesbadener Eisenbahngesellschaft“) - WEG

Lahn Valley Railway („Lahntalbahn“) - LTB

When „brown phase“ starts the countries are united in the German Empire, so concessions for the four countries become meaningless. Only the right of passage for Frankfurt's special track stays in effect, although it can't be bought any more then since the privates close at start of „brown phase“.

8.4.2 Laying or upgrading track

A company may lay or upgrade exactly one track tile per OR. It can skip this part of an OR. There are basically three kinds of hexes: hexes with cities are symbolized by one or more white-coloured circles, hexes with small towns are symbolized by block dots, hexes without cities or towns don't have such circles or dots. Only track tiles with matching symbols may be placed on hexes.

An empty hex gets a yellow track tile first. After „green phase“ has started by purchase of the first step-3-train, yellow track tiles already on the map can be upgraded to green track tiles. After „brown phase“ has started by purchase of the first step-5-train, green track tiles already on the map can be upgraded to brown track tiles. The green printed hex of Bad Homburg counts as a green tile for this purpose, it can directly get the special brown Bad Homburg tile after start of „brown phase“. Some hexes show symbols representing costs for laying track onto this hex. The track laying company must pay these costs from its treasury. There are no costs for upgrading the hex later in the game. The costs are listed in chapter 11.2 . Furthermore the following rules must be respected:

- To lay or upgrade track on a hex, a company must be able to reach at least any piece of track on the hex after the track lay by following a route there from an own station marker. Note that you still can't pass cities, where all circles are occupied by station markers of other companies (see 8.4.4).
- Eventually the company must purchase a concession before laying track (see 5.2 und 8.4.1).
- When you upgrade a yellow track tile to green or a green track tile to brown all existing track on the hex must be retained and station markers stay on the hex (if any).
- You must not build track that would lead outside the hex grid of the map.

- Track can only lead to a red off-board hex, if there is a track entrance (see map).
- You must not build track that would lead into a blue barrier, which represent Rhine river, Main estuary or Eder valley (where today is a reservoir called „Lake Eder“).
- During „yellow phase“ and „green phase“, only the FLB private is allowed to lay or upgrade track in the two Frankfurt hexes. Always use the special „F-tiles“ for these two hexes.
- There is one special track tile per colour with a printed „M“ on them for the Mainz hex. There are 4 yellow, 4 green and 3 brown special track tiles with printed „KGD“ on them that have to be used for the hexes with the cities of Kassel, Gießen and Darmstadt.
- The Wetzlar hex counts as brown track, so track can't be placed there.
You must not build track that would lead into Wetzlar on one of the three sides without track.
- When certain privates are exchanged or closed, special track laying can happen (see 5.3).
Track tiles built that way don't need to be connected to other track.
- Track tiles are limited. When all track tiles of a kind are on the map, you can't lay it at the moment. But when a track tile is upgraded, it comes back to the track tile pool and is available again.

Additionally once per its OR, a company may build one bridge over Rhine river or Main estuary, as long as this extends the company's possible routes. Such a bridge costs 60 Mark.

Up to 2 bridges, which are built by the private Building Company Rhine Bridges, are free of costs.

A bridge only can be built between cities or towns, so there are 6 possible positions:

Mainz - Wiesbaden, Mainz - Groß-Gerau, Mainz - Rüsselsheim,

Wiesbaden - Rüsselsheim, Worms - Bensheim, Bingen - Rüdelsheim.

Bridges count as normal track between the cities or towns for all track or route purposes.

After being built, bridges stay on the map for the rest of the game.

A company doesn't need to have a concession to build a bridge to another country.

But a company needs a concession of course to reach the other city, when using the bridge as track.

8.4.3 Obligated track routes

Every company has an obligated track route (see 11.1 or company sheets). A company's track lay or upgrade can complete such an obligated track route for any company. An obligation is completed, when track makes the full route and the route isn't blocked by station markers of other companies anywhere at this time (see 8.4.4). Furthermore, if the route goes through other countries but the company's home country, the company will need the corresponding concessions to claim completed obligation during „yellow phase“ and „green phase“, otherwise the company will have to wait until purchase of these concessions or until „brown phase“. To fulfill its obligation the company doesn't need to be able to use the obligated route with a train during income calculation. It will be enough if the route exists and theoretically is usable for an imaginary unlimited train. Note that the order of cities can be mandatory for the obligated track route. When its obligated track route is completed, the company gets the second half of its starting treasury (see 7.3). Companies that are founded after the start of „brown phase“ ignore their obligation and get their full treasury when floated (see 7.3).

8.4.4 Placing station markers

Once per OR, a company may place a station marker on a free city circle. It has three such markers, one of them is the home station marker. That home station marker is placed automatically for free in the company's home city at the start of the companies first OR. That doesn't count as a marker placement already, so the company is allowed to place another station marker in its first OR. Nine of the ten companys have two more station markers, one of them costs 40 M to place, the other one costs 100 M to place. Only FHB has two station markers for free, its home station marker in Fulda and a special marker in Hanau, so it has only one more to place for 40 M and none for 100 M.

To place a station marker in a city a company must be able to reach the city circle by following a route there from an already placed own station marker. The route still must not pass blocked cities as described earlier. It is not allowed not place any station markers in Frankfurt or Wetzlar or in small towns or in red off-board locations. If a city is the home station of a company that hasn't operated yet, one city circle must be kept free, until the company has place its home station marker

there. This rule also applies for Hanau until FHB's special marker is placed there. Yellow city tiles have one city circle, green city tiles and most of the brown city tiles have two city circles, some brown city tiles have three city circles. When all circles in a city are occupied by station markers, no more station markers can be placed there and other companies still can reach this city, but they can't pass this city any more, neither for track building, nor for completing the obligated track route, nor for placing station markers, nor for running trains. To place a station marker outside of a company's home country, the company needs the corresponding concession (see 8.4.1).

8.4.5 Operating trains and calculation of revenue

Now the company calculates its revenue by operating its trains imaginarily on the map. Note that this part of an OR is done before purchase of trains, so it is possible (f.e. in the company's first OR), that the company eventually doesn't have a train yet at this point and therefore has a revenue of 0. Trains, that will be purchased in an OR, will operate in the following OR, not in this OR.

To calculate the revenue every train of the company is assigned to an existing route on a map. Different trains of the company must use completely different routes in the same OR. Not even a very short piece of track is allowed to be used twice by the company in the same OR. On the other hand a station is allowed to be part of routes of different trains, as long as the trains use different tracks to get into and/or out of the station. Every train's route must have a station marker of the company somewhere in the route. Bridges can be used for routes, if both cities at both ends of the bridge have a station value. The number of cities and small towns that can be in a train's route depends on the kind of the train (see 8.4.6). A red off-board area counts as a city has to be a start or end of a route. The same train can't have any city or red off-board area more than once in the same route. This rule also applies for the two different stations of Frankfurt since both count as Frankfurt. The special track between the two stations of Frankfurt only can be used by companies which have purchased the concession for city passage of Frankfurt from the owner of the Frankfurt Local private. When a train uses Frankfurt's special track, it counts the two stations of Frankfurt as one city station and uses Frankfurt's station value only once (the higher value if these are different). Note that companies can't purchase the concession for city passage of Frankfurt any more during „brown phase“, because the remaining privates close at the start of „brown phase“. All other concessions become unimportant in „brown phase“, all companies may operate anywhere then.

A train's revenue is the sum of all station values in its route. A company's revenue is the sum of all the revenues of its trains. Wetzlar's value is 20 in „yellow“, 30 in „green“, 40 in „brown“. All red off-board areas have a table in their hex showing their values in „yellow“, „green“ and „brown“.

A train with a route from one red off-board area to another one operates a so called transit route and gets a bonus on its revenue for the transit. To calculate the bonus all of the red off-board areas are assigned to one of the four cardinal directions: *Hannover* and *Ostwestfalen* are northern stations, *Thuringen* and *Franken* are eastern stations, *Baden* and *Pfalz* are southern stations, *Rhineland* and *Suedwestfalen* are western stations. The bonus for a transit route is the sum of both transit values of the two included red off-board areas, each adding the value for the other one's cardinal direction.

Example: A transit route from Rhineland to Franken grants a bonus of 60, because Rheinland grants 40 for a transit route to the east and Franken grants 20 for a transit route to the west.

The company's director always is obligated to calculate the optimized revenue for the company. But he is not forced to do other actions in the OR to improve the revenue, f.e. purchase a concession.

The revenue is paid out by the bank. A company has two options what to do with its revenue. It can keep the money for its treasury moving its share value marker one box to left, or it pays out the money as dividends to its share holders (a tenth of revenue per 10% in shares) moving its share value marker one box to right. When paying out the dividends, dividends for shares in init, in pool or for a reserved share, whose private hasn't been exchanged yet, stays in the bank. A company that doesn't have a train in this part of an OR can't pay out a dividend, so the share value marker moves one box to left. A share value marker in the left or right column, which shall move one more box in this direction, follows the arrow instead, but it doesn't move beyond the highest or lowest box.

8.4.6 Types of trains and purchase of trains

The final part of a company's OR is train purchasing. Now the company can purchase one or more trains. From game start on, each company is allowed to possess up to four trains at the same time, this limit decreases to three trains for every company, when the first level-4-train is purchased by any company, and this limit decreases to two trains for every company, when the first level-5-train is purchased by any company. An operating company cannot sell a train in this phase or voluntarily drop one to get room for a new train, because this phase is for purchasing trains, not for selling trains. So when a company is at the current train limit, it can't purchase another train, even if it would get new room for trains by starting a new train phase. If the company doesn't have any train in this phase, it will be forced to purchase one in this phase from any source (see 8.4.7).

The company can purchase a train either from the bank or from another company. To purchase a train from the bank the company pays the train's value to the bank from its treasury. To purchase a train from another company, which is allowed during the whole game, the train's price is negotiated by the two directors (which can be the same player) and paid from the buying treasury to the selling one, but the price must be at least 1 Mark per train. In both cases the money must be completely be available in the purchasing company's treasury, if the company already has a train. If the company is forced to purchase a train, because it doesn't have one yet, the director can choose any available train - it doesn't need to be the cheapest one - and the train's price may exceed the company's treasury. In that case the company's director will face an emergency money raising (see 8.4.7).

There are two types of trains and five different train levels. The two types are „normal trains“ and „plus-trains“. The train levels go from level 2 to level 6. For a normal train its level says, how many stations (or less) the train's route may include, no matter if these are cities or small towns. A plus-train can add as many small towns (or less) as its level says. (f.e.: a 3+3-train's route can include up to 3 cities and 3 small towns or 2 cities and up to 4 small towns, etc.)

There are 9 level-2-trains, 6 level-3-trains, 5 level-4-trains, 4 level-5-trains, 11 level-6-trains.

When purchasing a train from the bank, a company must decide whether the train will be a normal one or a plus-train. Note, that plus-trains are more expensive. The difference is the train level multiplied by 10 Mark for level 2 or 3 and multiplied by 20 Mark for the other levels. So prices are: 2: 80, 2+2: 100, 3: 150, 3+3: 180, 4: 320, 4+4: 400, 5: 450, 5+5: 550, 6: 600, 6+6: 720. After purchased from the bank, trains never change their type again, even when traded.

The trains in the bank must be purchased in order, so first all nine level-2-trains before the first level-3-train, and so on.

Each time when the first train of a new train level is purchased, a new train phase begins (see 11.5). At start of phase 3 and phase 5, new track tiles become available (see 8.4.2.). At start of phase 4, all level-2-trains (no matter what type) are taken out of the game without compensation, at start of phase 6, all level-3-trains (no matter what type) are taken out of the game without compensation. Because of this effect it is possible, that companies don't have a train any more and will be forced to purchase a train during their next OR. Furthermore, at start of phase 4 and phase 5, train limits decrease (see 11.5), so it is possible, that companies suddenly have too many trains (f.e. 3 trains when the first level-5-train is purchased). In this case such a company skips a train to the bank by director's choice without compensation. A train skipped to the bank can be purchased again as a normal train or as a plus-train by any company, but the train doesn't need to be bought before higher-leveled trains. To remember this, skipped trains are put into the bank pool instead of the train boxes. You can find an overview of the train level effects in appendix 11.5.

8.4.7 Forced train purchase, emergency money raising and bankruptcy of a player

A company that doesn't have a train in its OR is forced to purchase one during the respective part of its OR. The company's director chooses, whether to purchase an available train of any type from the bank paying the train value to the bank or to purchase a train from another company – if that company's director agrees - for at least 1 Mark. Note, that in the latter case there is no limit to the price the two directors negotiate.

Such a forced train purchase is the only way to purchase a train for more money than the company's treasury has available. In this case the company has to do an emergency money raising: The director of the purchasing company must pay the difference from his private cash to make the train purchase possible for his company. Now: If the director himself doesn't have enough cash to pay the difference, he will be forced to sell several of his shares one after another until he has made enough cash. It is this player's choice which shares to sell in which order, but this special selling action stops, when the needed cash is made. All selling rules as described in chapter 7.1 apply. Share values of all companies whose shares are sold during emergency money raising decrease by one share value box as during stock rounds. This can even change the order of companies. In the worst case the director could have sold all the shares the director is allowed to sell, but still there is not enough player's cash to pay the difference. If this happens, the player will be bankrupt. The game will end immediatly and the bankrupt player will finish the game in the last place. The other players now calculate their wealth to determine the winner (see chapter 4).

The special selling action during an emergency money raising can result in changes of directorship. These are carried out as described in chapter 7.4, but no before the emergency money raising is finished. This rule guarantees, that the operating director himself suffers all the consequences of the emergency money raising, even if he loses the company with the new train to another player during the process.

8.5 Conclusion of an operation round

After the last company of the current OR has operated, exchange of some privates (Building Company Eder Valley Dam, Taunus Railway or Odenwald Railway) eventually could be forced (see 5.3). Then the next OR or a new SR depending on train phase begins (see 8.1). As long as the game currently is in „green phase“, don't forget the intermediate round for the option to exchange certain privates if wished by their owners. The intermediate round also happens before a stock round.

At the end of a full set of a stock round and its following operation rounds the game will end, if during the set the bank has run out of money at least once (0 Mark counts as running out of money). Of course the bank will continue to pay out dividends until the end of the game. At the end of a game, players calculate their wealth to determine the winner as described in chapter 4.

9. Rule variants

Alternatively, the initial auction can follow the procedure of games like *1830*, *1856*, *1870*, etc., in case all players wish to do so. In initial turn order players either can buy the private on the top for face value or place a bid on a private, which is later in private order, or pass. Placing a bid on a private, which is not on top of private order must be at least 5 Mark more than face value or the highest bid so far for this private and all bids must be multiples of 5. When the private on top is bought and the next private has got at least one bid, an auction for this next private starts involving players who have bid on this next private. So if only one player has a bid on this private, this player will get it for the price of his bid. This procedure continues until a private comes to the top, which hasn't got a bid so far, or until all privates are sold. If you use this system for initial auction, player order won't be set newly after initial auction, but the first SR will start with the player who is sitting behind the last player that has bought a private regularly for face value.

If in this system of initial auction all players pass in a row without buying Homburg's Railway, its price will decrease to 15 Mark, then to 10 Mark, then to 5 Mark, then to 0 Mark. The player, whose turn it is, will then be obligated to buy Homburg's Railway für 0 Mark.

If in this system of initial auction all players pass on a later private that hasn't got a bid, the initial auction will be stopped for a short OR, because some privates already have been bought. If „Frankfurt's Local Railway Co.“ already has been bought it can lay track in a Frankfurt hex in this short OR. The privates pay their income to their owners now. Afterwards, continue initial auction with the private that hasn't been bought before.

In case all players want to do so, player order can be reorganized like in 1844 Switzerland or in the 1822 games at the end of an SR for the next SR. In this variant the player with the most money after an SR will be player 1 for the next SR, the player with the second most money player 2 and so on. In case of ties the old player order will be the tie-breaker.

Author's note: The variant to give the priority deal to the player with the most money after an SR but to keep player order generally has been tested and this hasn't worked for the game, so this idea can't be recommended.

In case all players wish so, initial auction can involve one more private, which is described below. Players starting money is increased by additional 10 Mark from the bank in this case.

This additional private will be auctioned as the second private of the initial auction private order (after Homburg's Railway and before Building Company Rhine Bridges):

2. Building Company Seidler und Siebrecht face value: 30, income: 10

special ability: It closes at start of brown phase at the latest.

During yellow phase and green phase any company whose director owns this private is allowed to ignore the complete cost of one track lay (but not a bridge over Rhine river) once per game.

By using this special ability one time the private closes before start of brown phase.

Please note that addition of this paper hasn't been part of play testing.

10. Historical notes

Directly after the first run of Germany's first train (the „Eagle“) from Nuremberg to Fuerth in 1835, in all the small German states plannings of railway tracks began and financing companies for these projects were founded. But since there wasn't a united German state yet, these early plans were made only on the level of these regional small states. The Kingdom of Bavaria for example developed its internal railways quite fast. On the other hand, discussions about interregional routes started soon, especially in the small states located in central Germany. There were several small states in the area we call Hesse today, each of them planning and constructing independently and sometimes even in competition with its neighbours.

In 1836, only one year after the „Eagle's first run“, a first railway comitee was founded in Frankfurt, which was a city state at this time and already a local metropolis, to plan a track from Frankfurt via Hoechst at the Taunus mountains to Wiesbaden, the capital of the Duchy of Nassau, using Nassau's territory, and in Wiesbaden you could pass the Rhine river to Mainz. This so-called „Taunusrailway“ was strongly supported by the duke of Nassau, because he was worried, other plannings could result in a route from Frankfurt to Mainz on the southside of the Main river using the territory of Hesse-Darmstadt before a realization of Nassau's track, what could decrease any potential of railway construction in Nassau at all. Parallely on January 2nd of 1836, another comitee had been founded in Frankfurt, which discussed the idea of connecting all the adjacent big cities Mainz, Darmstadt and Gießen to Frankfurt. So the „Taunusrailway“ was planned and constructed faster and became the first finished railway in Hesse. Later, this route was extened to Ruedesheim and then even to Koblenz in the Rhineland alongside all of Nassau's eastern shore of the Rhine river. The „Taunusrailway“ evolved into „Wiesbaden's Railway Company“. But this was Nassau's only railway project for a long time. Further plans for a track along Lahn river from the Prussian city of Wetzlar to Lahn's estuary in Lahnstein or a route along Dill river to the old Oranian capital Dillenburg and then on to southern Westfalia were realized much later. And these three routes were the only ones apart from some short local tracks in Nassau until the construction of a high-speed route from Frankfurt to Cologne for the German trains called „Intercity Express (ICE)“ in the 1990s.

Frankfurt's comitees became the impetus to railroad construction in Hesse. But at the same time, Frankfurt refused to allow routes, which would lead directly through the whole city, to protect the interests of Frankfurt's conveyances. Even today Frankfurt's Central Station still is a terminal station. Concepts for an underground passage station, which have been planned in the 20th century have been iced since several incidents around the project of „Stuttgart 21“. On the other hand, Frankfurt's railway concepts forced the surrounding sovereigns and their assemblies to discuss their own goals for railway construction and its concessions in their countries as well. Especially the Grand Duchy of Hesse-Darmstadt was very interested in a connection of its capital to big adjacent cities like Frankfurt and Mannheim through the territory of its core area „Starkenburg“, which is southern Hesse today. Therefore an interregional treaty with Frankfurt and the Grand Duchy of Baden was signed in 1838 already. But Darmstadt's railway company, which was founded for that reason, ran out of money within three years and had to be closed in 1841. So construction of a so-called Main-Neckar-Railway became a governmental business now. Darmstadt's government followed a long-termed vision to extend the route from Mannheim to Frankfurt in the direction of Giessen, which was the second major city of Hesse-Darmstadt administrating the province called Upper Hesse. They even hoped for a northern extension to Kassel. Because of these ideas and starting track constructions in Bavaria, Saxonia and Prussia questions of economic efficiency became less important. It was realized, that lack of such routes could become a bitter disadvantage for a region. Hesse-Darmstadt already had to swallow a bitter pill with the fact, that Mainz, which was administrating the province of Rhinish Hesse, was connected to Frankfurt via Nassau's territory. In Darmstadt this experience resulted in big sympathies for all route ideas that made any sense, although they couldn't build everything at the same time. A direct route from Mainz to Darmstadt and a route to Offenbach were strongly demanded, even a track through the mountain terrain called Odenwald was favoured by the represents of Starkenburg and represents of Upper Hesse wanted a route from Giessen to the peripheral Vogelsberg (= bird mountain) area. The Vogelsberg Railway (also called Upper Hesse Railway) starts the game in the Vogelsberg hex, not in Giessen as historically, to keep game balance. But Hesse-Darmstadt's most successful and longest lasting company was Hesse's Louis Railway, an offspring of Louis Railway in Palatia. Hesse's Louis Railway started in Mainz and not only realized demanded routes in Starkenburg, but also completed a full route on the western shore of Rhine river from Palatia's industrial city Ludwigshafen via Mainz to Koblenz in the Rhineland where Mosel river meets Rhine river. Together Hesse's Louis Railway and Main-Neckar-Railway turned Hesse-Darmstadt into the centre of railroad development in the region of Hesse. During this process, the small county of Bad Homburg in the east of the Taunus mountains was ignored though. Bad Homburg wasn't connected by railroad to the Frankfurt area before the German Empire was founded.

In contrast to Hesse-Darmstadt, Hesse-Kassel had far more problems developing railroad in its territory. The first route across Kassel's territory was built in 1854 only by initiative of Frankfurt's comitees from Frankfurt-East along the northern side of Main river to Frankonia via Hanau. This „Frankfurt & Hanau railway“ was absorbed by other companies later. The Elector of Hesse-Kassel had big plans to make his capital an important centre of railway tracks for connection of southern German countries to the north and for a connection between the eastern and western parts of Prussia. But these big plans failed in early stages already because of many reasons. There were bitter conflicts in Hesse-Kassel's parliament involving several groups of members. One group wanted to see a Main & Weser railway to the south via Marburg and Giessen, another group rooted for a route using only the own territory from Kassel along the rivers Fulda and Kinzig to Fulda city and Hanau. And there was even a group around assembly members Baehr and von Ochs, who absolutly refused any railroad, which resulted in a famous caricature in 1844, which is the cover picture of this game. Thus, after many difficulties at the start, a regional route was realized in northern Hesse-Kassel including a connection to the kingdom of Hannover finally in 1856. Later the supporters of a route to the south via Marburg won the debate, because this version of a Main & Weser railway was prejudged by Darmstadt's track building to Gießen. The loser was the city of Fulda in the east becoming periphery now. So in Fulda plans started for a route from Thuringia to Kassel via Fulda,

but track building didn't go farther than to Bad Hersfeld („Fulda & Hersfeld railway“), after that Hesse-Kassel was annexed by Prussia as a result of the Prussian-Austrian war. The long-desired eastern connection of the cities of Fulda and Kassel was realized a long time later, when the old country of Hesse-Kassel didn't exist any more. The two small companies in eastern Hesse-Kassel and their special history have been combined to one company in this game for balancing reasons. In an earlier version they were two small companies, which merged during the game. Further projects with less acknowledgements in Hesse-Kassel led from Marburg to the west via Biedenkopf and along Eder river to the north heading for the Principality of Waldeck. This track from Marburg to Korbach via Frankenberg („Lahn & Eder railway“) shows a wonderful landscape and originally was my first motivation in 2001 to make this game. Unfortunately this Lahn & Eder railway is among the ideas, which didn't make the cuts during developmental process. But I like the fact, that in reality this track was reopened for railroading a few years ago.

One certain company called Waldeck's Local Railway has stayed in the game since the beginning despite all the changes involving it. This is the only non historic company in the game, it never has existed in the way the game presents it. It is the author's appreciation to this nice small country called Waldecker, which has been absorbed by Hesse in 1929, so nearly nobody knows it outside of Waldeck any more today. But Waldeck has several routes going in a star's formation from the capital Korbach, of course these are not very important outside of the region. Although Waldeck didn't play a roll in early railway building during the time of the game, it shouldn't be forgotten in the game, because it is part of Hesse today. Furthermore Waldeck's Local Railway shall have the effect for the game, that the map's northwestern part doesn't become too less interesting.

It is a fact for all of the old countries in Hesse, that railway development has been difficult mainly for financial reasons until 1866 or 1871, when they were integrated into the new German Empire. This applies especially in the rough terrain in the north of Main river. But as soon as planning was concluded and financial aspect were secured, projects were realized quickly and professionally, mainly with the help of very capable building companies, which included the input of experienced Belgian experts. The biggest and most popular building company Seidler & Siebrecht has been covered in a little game variant as described in section 9.

Of course this short overview cannot cover all the historic details that could be told about railways in Hesse. Anyone, who wants to get more details and facts, should read Ludwig Brake's book „The first railways in Hesse“, published in Wiesbaden by the Historic Commission of Nassau in 1991. The importance of this book for the development of this game can't be valued highly enough.

notes by Karsten Ockenfels, April 2016

11. Overviews and tables

11.1 Companies

Duchy of Nassau

Wiesbaden's Railway Company („Wiesbadener Eisenbahngesellschaft“): game colour: green
home station: Wiesbaden track obligation: Wiesbaden-Rüdesheim-Rheinland **WEG**
One 10%-share is reserved as exchange share for Taunus Railway.

Lahn Valley Railway („Lahntalbahn“): game colour: orange
home station: Limburg track obligation: Rheinland-Limburg-Wetzlar **LTB**

Grand duchy of Hessen-Darmstadt

Main & Neckar Railway („Main-Neckar-Bahn“): game colour: blue
home station: Bensheim track obligation: Baden-Bensheim-Darmstadt-Frankfurt **MNB**

Starkenburger's Railway („Starkenburger Bahn“): game colour: grey
home station: Darmstadt track obligation: all three hexes of Odenwald any way **SB**
One 10%-share is reserved as exchange share for Odenwald Railway.

Hessian Louis Railway („Hessische Ludwigseisenbahn“): game colour: white
home station: Mainz track obligation: Rheinland-Bingen-Mainz-Pfalz **HLB**

Vogelsberg Railway („Vogelsbergbahn“, *Engl.: Bird Mountain Railway*): game colour: brown
home station: Alsfeld track obligation: Alsfeld-Gießen **VB**

Electorate of Hessen-Kassel

Main & Weser Railway („Main-Weser-Bahn“): game colour: red
home station: Marburg track obligation: Marburg-Gießen-Frankfurt **MWB**

Friedrich Wilhelm Northern Railway („Friedrich-Wilhelm-Nordbahn“): game colour: violett
home station: Kassel track obligation: Fritzlar-Kassel-Bad Karlshafen **FWN**

Fulda & Hanau Railways („Fulda-Hanauer Bahn“): game colour: pink
home station: Fulda track obligation: Fulda-Bad Hersfeld **FHB**

One 10%-share is reserved as exchange share for Frankfurt & Hanau Railway.
One station marker is for free in Hanau when doing the exchange. The other one costs 40 M.

Principality of Waldeck

Waldeck's Local Railway („Waldeck'sche Landesbahn“): game colour: yellow
home station: Korbach track obligation: Bad Wildungen-Korbach-Arolsen **WLB**
One 10%-share is reserved as exchange share for Building Company Eder Valley Dam.

11.2. Track building costs

When laying track companies face these costs out of their treasury:

river hex (wave symbol): 20 M ; mountain hex (triangel symbol): 40 M

both symbols (Biedenkopf, eventually Bad Wildungen): 60 M

bridge over Rhine river or over estuary of Main river: 60 M

buying a concession: 40 Mark (This is not exactly track building, but follows the same idea.)

11.3 Transit routes

Hannover (N): S:+40, N:+20, O:+20, W:+20

Ostwestfalen (N): S:+40, N:+10, O:+30, W:+10

Suedwestfalen (W): S:+20, N:+20, O:+20, W:+20

Rheinland (W): S:+10, N:+60, O:+40, W:+20

Thuringen (O): S:+30, N:+10, O:+20, W:+30

Franken (O): S:+10, N:+40, O:+30, W:+20

Pfalz (S): S:+10, N:+40, O:+30, W:+20

Baden (S): S:+10, N:+40, O:+20, W:+30

(engl.: Eastern Westfalia)

(engl.: Southern Westfalia)

(engl.: Rhineland)

(engl.: Turingia)

(engl.: Francia)

(engl.: Palatia)

11.4 Trains

<i>normal train</i>	<i>cost</i>	<i>combi train</i>	<i>cost</i>	<i>events when first such train bought</i>
2	80 M	2+	100 M	train limit 4
3	150 M	3+	180 M	start green phase
4	320 M	4+	400 M	eliminate trains 2 and 2+, train limit 3
5	450 M	5+	550 M	start brown phase, train limit 2
6	600 M	6+	720 M	eliminate trains 3 and 3+

11.5 Game phases

phase	starts when...	available track	train limit	number of OR after SR in set	eliminate trains	other effects
1	initial auction	yellow (FLB only)	4	1 (privates only)		
2	first SR	yellow	4	1		see rules for first SR
3	purchase of first train 3 or 3+	yellow and green	4	2		exchange of privates possible
4	purchase of first train 4 or 4+	yellow and green	3	2	2 and 2+	exchange of privates possible
5	purchase of first train 5 or 5+	yellow and green and brown	2	3		close privates, close concessions, keep Frankfurt pass. evtl. build Homburg, companies may lay track in Frankfurt, companies founded afterwards get full capitalization when floating
6	purchase of first train 6 or 6+	yellow and green and brown	2	3	3 and 3+	